Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/03/2023
PI Name:	Hogan, John Ph.D.		
Project Title:	Investigating the Physiology and Fitness of an Exoelectrogenic Microorganism Under Microgravity Conditions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
PI Email:	John.A.Hogan@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-0152
Organization Name:	NASA Ames Research Center		
PI Address 1:	Code SCB, MS 239-15		
PI Address 2:			
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-0001	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	09/30/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2019 per F. Hernandez/ARC (Ed., 3/11/19) NOTE: Extended to 1/31/2019 per F. Hernandez/ARC (Ed., 8/21/18) NOTE: Extended to 4/30/2018 per F. Hernandez/ARC (Ed., 5/2/17)		
Key Personnel Changes/Previous PI:	September 2015 report: During the course of this performance period, the Co-PIs Dr. Lily Chao and Dr. Masood Hadi have transitioned and are no longer involved with the project. 2016: Dr. Michael Dougherty has joined the team as a co-investigator.		
COI Name (Institution):	Dougherty, Michael Ph.D. (Wyle/NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary objective of this proposal is to gain fundamental understanding of the physiology of a microorganism (exoelectrogen) that is capable of respiring on solid electron acceptors such as metal oxides during spaceflight. These specialized organisms are used in bioelectrochemical systems, which have the potential to conduct life support processes such as wastewater treatment as well as yield a wide array of space-relevant products such as biofuels, food products, and therapeutics. The ability to respire on solid electron acceptors is dependent on a process called extracellular electron transport (EET) and is often biofilm-dependent. We hypothesize that the performance of exoelectrogens, particularly when respiring on solid electron acceptors, will be altered during spaceflight. Therefore, we propose to characterize the growth and biofilm development of a model exoelectrogen, Shewanella oneidensis MR-1 under EET-respiring spaceflight conditions. Additionally, we will use whole-genome expression and competitive mutant fitness (i.e., omic approaches) profiling to identify genes that confer a selective advantage or disadvantage for cell growth and biofilm development under EET-respiring conditions during spaceflight. Together, we anticipate that the results of these experiments will not only help guide future research in strain development for bioelectrochemical systems but also provide further insight into the impact of microgravity and the space environment on microbial growth and physiology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect to gain a fundamental insight into the impact of microgravity and the space environment on microbial growth and physiology of a model exoelectrogenic organism. We anticipate that the results of these experiments will help guide future research in strain development for bioelectrochemical systems for use both in space and terrestrial applications. These organisms are utilized in microbial fuel cells, and have the potential to improve terrestrial wastewater treatment systems by off-setting power requirements and producing less bacterial sludge. Such systems may also be utilized for advancing sanitation in developing countries. Also, a deeper understanding of the processes in extracellular electron transfer could be gained, which can result in improved engineering of exoelectrogens to increase overall performance.
Task Progress:	[Ed. Note, November 2023: Compiled from final progress reported by Principal Investigator (PI).] The Micro-12 flight experiment was launched on SpaceX-15 and completed during berthing on the International Space Station. The goal of this experiment was to understand the effects of spaceflight and microgravity on the physiology of the model exoelectrogen Shewanella oneidensis MR-1. The BioServe Fluid Processing Apparatus (FPA) and Group Activation Pack (GAP) hardware systems were used for both flight and ground control tests. Under spaceflight conditions, extracellular electron transfer (EET) rates were found to be significantly increased on insoluble substrates, while biofilm development appeared to be unchanged under the conditions tested; these processes are critical for microbial-assisted bioelectrochemical systems. Additionally, RNAseq analysis, proteomic profiling, and competitive mutant fitness profiling were performed to gain further understanding of microbial physiology under EET-respiring conditions during spaceflight. Overall, the results of the Micro-12 project support the idea that Shewanella oneidensis MR-1, in particular – and exoelectrogens in general – could be useful chassis organisms for synthetic biology applications using microbial bioelectrochemical systems. These findings will assist bioengineering and synthetic biology development efforts harnessing the unique capabilities of exoelectrogens for life support and in situ resource utilization. • The final dataset for the Shewanella mutant fitness library was provided by our collaborators at Lawrence Berkeley National Laboratory. • An abstract was submitted to the American Society for Gravitational and Space Research (ASGSR) for an oral presentation for the November 2020 conference in Denver, CO.
Bibliography Type:	Description: (Last Updated: 11/03/2023)
Abstracts for Journals and Proceedings	Dougherty M, Deutschbauer A, Ball N, Price M, Ray J, Karouia F, Arkin A, Hogan J. "Results of the Micro-12 Flight Experiment: Effects of microgravity on Shewanella oneidensis MR-1." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
NASA Technical Documents	Dougherty M, Deutschbauer A, Ball N, Karouia F, Price M, Ray J, Arkin A, Hogan J. "Results of the Micro-12 Flight Experiment: Effects of microgravity on Shewanella oneidensis MR-1." Moffett Field, California : NASA Ames Research Center, 2019. NASA/CP-2019-0033397. https://ntrs.nasa.gov/citations/20190033397 , Nov-2019